



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

(An Autonomous Institution Affiliated to Anna University, Chennai)

SATHYAMANGALAM - 638 401

Module Test - I - Apr - 2026

II Semester

22AI206/22AM206/22CS206/22IT206 & DIGITAL COMPUTER ELECTRONICS

1	(i)	A telemetry system converts decimal data into binary and then into octal for transmission. Engineers observe incorrect outputs when fractional values are processed. The system uses standard conversion techniques but suffers from truncation and grouping inconsistencies, affecting final transmission accuracy. Diagnose the cause of incorrect outputs and infer how binary conversion errors impact octal representation. (3 Marks - [An/M, 2])
2	(ii)	A calculator device internally processes numbers in binary but must display outputs in a user-friendly format. The display has limited space, and binary representation appears lengthy. The system designer must choose between octal and hexadecimal formats for output representation while ensuring accuracy and minimal processing overhead. Determine the most appropriate number system for display and justify the decision based on efficiency and usability. (2 Marks - [Ap/C, 1])
3	(i)	A system processes signals through encoder, comparator, and decoder modules. Incorrect sequencing leads to wrong outputs. Formulate the correct processing sequence and infer the impact of improper sequencing. (3 Marks - [An/P, 2])
4	(ii)	A control system receives multiple request signals and must encode them into binary form for processing. When multiple requests occur simultaneously, the system produces incorrect outputs due to lack of priority handling. The designer must improve the encoding mechanism. Determine the appropriate encoder type and justify the decision based on handling multiple inputs efficiently. (2 Marks - [An/F, 2])
5	(i)	A processing unit converts inputs from multiple number systems into binary, performs operations, and converts results back into original formats. Improper sequencing has caused incorrect outputs in previous tests. The system must ensure proper data flow and accuracy. Formulate a correct sequence for number system conversion and infer the impact of incorrect sequencing. (3 Marks - [E/C, 1])

6	(ii)	A program converts binary values into decimal but fails when fractional values are present, producing incomplete results. The integer portion is processed correctly using positional weights, but the fractional portion is ignored. The developer must identify missing logic affecting correctness. Identify the error in the program and determine its effect on output accuracy. (2 Marks - [Ap/C, 1])
7	(i)	A comparator circuit incorrectly indicates equality for unequal inputs due to improper handling of higher-order bits and cascading logic errors. Diagnose the issue and infer how it affects comparison accuracy. (3 Marks - [C/M, 2])
8	(ii)	A decoder used in a display system activates more than one output for a given input due to incorrect logic design, causing display errors. Identify the fault and determine its effect on system functionality. (2 Marks - [Ap/M, 2])
9	(i)	A digital system integrates signed number representation into arithmetic operations and must ensure consistency and accuracy across all computations. Design a consistent model for signed number representation and justify its effectiveness. (5 Marks - [E/P, 2])
10	(i)	A system designer evaluates multiplexers and demultiplexers for scalable data routing applications. Assess trade-offs and recommend an optimal design approach. (5 Marks - [E/P, 2])
11	(i)	A system computes $(-4) + (-2)$ using 4-bit 2's complement representation and must verify the correctness of the result within range limits. Determine the result using 2's complement and justify the steps logically. (5 Marks - [E/C, 2])
12	(i)	A 4:1 multiplexer must be implemented using Boolean logic to correctly select one input based on select lines. Analyse the Boolean expression and justify logic. (5 Marks - [An/P, 2])
13	(i)	A system designer evaluates using basic gates versus standard combinational modules for building scalable circuits. Assess trade-offs and recommend an optimal design approach. (5 Marks - [An/C, 2])
14	(i)	A system designer evaluates ALU architectures for efficient and scalable processing. Assess trade-offs and recommend optimal design. (5 Marks - [E/P, 2])

15	(i)	A Boolean function $F = A'B + AB'$ must be implemented and verified for correctness using logic gates and truth table analysis. Determine the circuit implementation and justify its behavior logically. (5 Marks - [Ap/P, 2])
16	(i)	A processor integrates an ALU combining arithmetic and logical units controlled by instruction signals. Design a coherent ALU model and justify effectiveness. (5 Marks - [E/P, 2])
17	(i)	A digital circuit designer is selecting between a latch and a flip flop for a system that requires controlled data storage only at specific time intervals. The system operates in a noisy environment where inputs may fluctuate frequently. Establish why a flip flop is preferred over a latch in this case. (3 Marks - [Ap/P, 2])
18	(ii)	A D flip flop receives a data input that changes continuously, but the output updates only when a clock pulse arrives. Determine the reason for this controlled behavior. (2 Marks - [U/C, 1])
19	(i)	A counter circuit used in a system begins to produce incorrect outputs due to delay in signal propagation between flip flops in a ripple configuration. Relate this issue to the type of counter used and suggest the improvement. (3 Marks - [An/C, 2])
20	(ii)	A 4-bit binary counter is used to count pulses from a sensor in an automated system. Identify the maximum count value before it resets. (2 Marks - [An/C, 1])
21	(i)	A latch changes output whenever the input changes, causing instability in a noisy environment. Relate this behavior to the need for flip flops. (3 Marks - [Ap/C, 2])
22	(ii)	An SR latch receives inputs $S = 1$ and $R = 1$ simultaneously. Identify the condition produced. (2 Marks - [Ap/C, 1])
23	(i)	A digital communication system requires conversion of parallel data into serial form for transmission over a single communication line. A shift register is used, where data bits are shifted one by one with each clock pulse until the entire data is transmitted. Establish how shift registers enable efficient data transmission in such systems. (3 Marks - [Ap/C, 2])

24	(ii)	A shift register initially contains the data 1011. With each clock pulse, the bits shift one position to the right, and a 0 is introduced at the leftmost position. Determine the output after one clock pulse. (2 Marks - [Ap/C, 1])
25	(i)	A digital system controlling an automated assembly line uses multiple flip flops to coordinate operations such as material feeding, processing, and packaging. Initially, the system faced issues where outputs changed at unpredictable times due to asynchronous input changes. To resolve this, a clock signal is introduced so that all flip flops update their states simultaneously at defined time intervals. Engineers also ensure that the clock frequency is chosen carefully to balance speed and reliability. Analyze how the introduction of a clock signal ensures synchronized operation in sequential circuits and prevents timing-related errors. (5 Marks - [U/C, 3])
26	(i)	In a microprocessor-based system, registers are used to temporarily store binary data during execution of instructions. Consider a system where a 4-bit register stores the value 1101. During the next clock cycle, this data is transferred to another register for further processing. Engineers ensure that proper control signals are applied so that data is retrieved from the first register and loaded into the second register without loss or corruption. The system uses a common bus architecture to facilitate data transfer between registers. Analyze the process of storing and retrieving binary data between registers and explain the role of control signals and clock in ensuring accurate data transfer. (5 Marks - [U/C, 3])
27	(i)	A digital communication system uses edge-triggered flip flops to store incoming data. The system is designed such that data is captured only at the rising edge of the clock signal. Engineers observe that even when input signals fluctuate between clock pulses, the stored data remains stable and unaffected. Evaluate how edge triggering improves data stability compared to level triggering in clocked circuits. (5 Marks - [An/C, 3])
28	(i)	A digital system consists of two 3-bit registers, Register A and Register B. Initially, Register A contains the binary value 101, and Register B contains 000. On a control signal, the content of Register A is transferred to Register B during the next clock pulse. Engineers verify that the transfer occurs simultaneously for all bits. Determine the contents of Register B after the transfer and explain how simultaneous data retrieval and storage occur. (5 Marks - [U/C, 3])
29	(i)	In a microprocessor system, registers are used to temporarily store intermediate results during arithmetic operations. A register receives binary data from an ALU and holds it until the next operation is performed. The system relies on clock signals to control when data is stored and transferred. Assess the importance of registers in data storage and processing within digital systems. (5 Marks - [An/C, 3])

30	(i)	A computer system is designed with a memory unit of 16 words, each word consisting of 8 bits. Engineers use address lines to uniquely select each memory location and a decoder circuit to activate the corresponding memory cell. When a specific binary address is applied, the decoder selects one memory location for read or write operation. The system must ensure that only one memory location is accessed at a time to avoid data corruption. Analyze how address decoding enables proper selection of memory locations in RAM and explain its role in read/write operations. (5 Marks - [An/C, 3])
31	(i)	A 4-bit register initially contains the binary value 1010. On the next clock pulse, the input data 0111 is applied to the register. The system is designed such that the register loads new data at every clock pulse. Determine the contents of the register after the clock pulse and explain the data loading process. (5 Marks - [An/P, 3])
32	(i)	In a microprocessor system, RAM is used to store instructions and data temporarily. During execution, the processor sends an address to the memory unit, and the corresponding data is retrieved. Engineers ensure that the memory is organized efficiently so that data access is fast and accurate. Assess the working of RAM in terms of storage, addressing, and data retrieval. (5 Marks - [Ap/C, 3])
33	(i)	A basic processor follows a structured instruction cycle where each instruction is processed in sequential steps controlled by clock pulses. The system includes registers, ALU, and memory interacting through control signals generated by the control unit. Each phase must occur in the correct order to avoid execution errors. The designer is analyzing how instruction flow is logically structured from start to completion. Determine the logical flow of instruction processing by interpreting the sequence of control steps, and justify how ordered execution ensures correct system behavior. (3 Marks - [U/P, 1])
34	(ii)	A processor executes instructions where each instruction requires a fixed number of clock cycles for completion. A given program consists of multiple instructions with varying cycle counts depending on operation type. The designer is analyzing execution time based on clock cycle distribution. The goal is to model total execution duration. Formulate the relationship between instruction count, clock cycles, and execution time, and determine how variations in cycle count affect total processing duration. (2 Marks - [Ap/C, 1])
35	(i)	A processor executes an instruction where the opcode specifies an arithmetic operation and the operand fields indicate register-based inputs. During decoding, the control unit must interpret the instruction and generate appropriate signals for ALU execution. However, timing constraints require precise sequencing of decoding and signal activation. The system uses a hardwired control unit with direct signal mapping. Engineers must outline the logical flow from instruction decoding to execution signal generation. Determine the logical sequence of operations from opcode interpretation to ALU control signal activation ensuring correct execution timing. (3 Marks - [Ap/C, 1])

36	(ii)	A simple processor simulator requires implementation of an instruction decoding module that interprets opcode values and triggers corresponding micro-operations. The system currently lacks structured logic for distinguishing instruction categories. Developers must design a decoding routine that maps opcodes to control actions. Formulate a pseudocode logic that ensures correct mapping of opcode values to corresponding control signals for execution. (2 Marks - [U/P, 1])
37	(i)	A system architect is evaluating processor performance under increasing workload where instruction complexity grows over time. The control unit must manage additional instruction formats and extended operation codes. The system must scale without significantly increasing execution time or control complexity. Trade-offs between simplicity and scalability become critical. Analyze the scalability challenges of the control unit under increasing instruction complexity, and recommend how control design choices influence system efficiency. (3 Marks - [An/C, 1])
38	(ii)	A processor integrates memory, registers, ALU, and control unit to execute instructions. The control unit generates signals that coordinate operations among these components. Efficient execution depends on proper interaction and timing of these elements. The system is being evaluated for operational coherence. Assess how the control unit ensures coordinated operation among system components and infer its impact on overall execution reliability. (2 Marks - [An/P, 1])
39	(i)	A microprocessor designed for IoT devices must balance decoding efficiency with limited hardware resources. The current combinational decoder is fast but consumes significant logic gates when supporting multiple instruction formats. Alternatively, a microprogrammed approach reduces hardware complexity but introduces execution delay. The system operates under strict energy and performance constraints. Engineers must evaluate the trade-offs between both decoding techniques. Analyze the trade-offs between combinational and microprogrammed decoding in this constrained system and infer the most efficient approach. (3 Marks - [An/C, 1])
40	(ii)	A computing system processes diverse workloads including arithmetic computations and I/O operations, but experiences inefficiencies due to generalized decoding logic. The instruction decoder does not differentiate execution paths effectively. Engineers consider restructuring the decoding unit to improve specialization. Assess how specialized decoding based on instruction types improves system performance and operational clarity. (2 Marks - [Ap/P, 1])
41	(i)	An industrial automation controller intermittently executes wrong instructions during operation. The system uses a sequential instruction flow where the PC increments after every fetch. However, during certain operations, the PC appears to skip addresses unexpectedly. Investigation shows that MAR receives correct addresses, but MDR occasionally loads outdated data. The control unit generates memory read signals without verifying memory readiness. Logs indicate that instruction alignment issues occur when accessing multi-byte instructions. Investigate the root causes of incorrect instruction execution by correlating PC behavior, MAR-MDR data transfer, and instruction alignment constraints, and determine the primary fault in the fetch mechanism. (5 Marks - [U/C, 2])

42	(i)	A 32-bit processor executes a sequence of arithmetic and branch instructions where operands are fetched from general-purpose registers and results are written back after ALU processing, while the control unit issues timed signals across clock cycles to coordinate register transfers, and a conditional branch depends on status flags generated during execution, and a temporary register is used to hold intermediate results for a multi-cycle multiplication, while instruction flow may alter based on branch resolution within the same execution window. Analyze the execution sequencing by determining the ordered micro-operations required to complete both the arithmetic and conditional branch instruction while reconciling how control signals synchronize data movement and influence control flow changes. (5 Marks - [An/C, 2])
43	(i)	A high-performance computing system processes large datasets and relies on deep memory hierarchies. During execution, instruction fetch delays increase significantly when instructions are not present in cache. The PC continues sequential updates, but MAR requests frequently go to slower main memory. MDR loading time increases, delaying IR updates. Engineers consider introducing prefetching or modifying bus organization. However, these changes may impact system complexity and resource utilization. Evaluate the trade-offs between improving instruction fetch performance using prefetching mechanisms and optimizing bus organization, considering their impact on PC sequencing, MAR-MDR operations, and overall system scalability. (5 Marks - [Ap/P, 2])
44	(i)	A high-performance processor pipeline executes arithmetic, logical, and branch instructions with overlapping stages, where execution latency increases due to multi-cycle instructions occupying ALU resources, and branch instructions introduce control hazards requiring pipeline stalls, while temporary registers are heavily utilized for intermediate storage, and design engineers must balance throughput against execution accuracy under varying workloads. Evaluate the execution mechanism by prioritizing trade-offs between pipeline efficiency and execution correctness while recommending strategies to optimize micro-operation sequencing under these constraints. (5 Marks - [Ap/P, 2])
45	(i)	A processor is executing an instruction that requires fetching an operand using indexed addressing where the base register contains 4000 and index register contains 120 while the offset is 30, and the system uses a unified bus for communication between ALU and memory. The control unit must coordinate address computation before initiating a memory read, while cache is checked prior to accessing main memory. Due to a recent workload spike, occasional wait states are introduced during memory access. The instruction involves a single operand stored in memory rather than registers. The processor must ensure correct sequencing of MAR, MDR, and IR during the operation. Analyze the sequence of operations required to determine how the effective address is computed and how the operand is fetched from memory while ensuring proper synchronization between components. (5 Marks - [An/C, 2])
46	(i)	A high-performance processor pipeline performs multiple instruction executions concurrently, but the write-back stage becomes a bottleneck when multiple instructions complete simultaneously. The system supports both register and memory write-back paths, and priority must be assigned dynamically. Increasing hardware resources can reduce delay but increases cost and power consumption. Designers must decide on an optimal approach. Evaluate the trade-offs involved in optimizing the write-back stage and recommend an efficient design strategy. (5 Marks - [Ap/C, 2])

47	(i)	A high-performance computing system must choose between register addressing and memory addressing for frequently executed arithmetic instructions. Register addressing provides faster access but limited storage, while memory addressing supports larger datasets but introduces latency due to cache and main memory interactions. The system executes multiple operand instructions requiring frequent data access. The control unit must balance speed and resource utilization under varying workloads. Engineers are evaluating which addressing mode yields better throughput under constrained conditions. Evaluate the trade-offs between register addressing and memory addressing in terms of performance, scalability, and system efficiency under these conditions. (5 Marks - [An/P, 2])
48	(i)	A processor completes an arithmetic instruction where the ALU produces a result that must be stored back into a destination register while also updating condition flags. The control unit activates specific write-enable signals, and the memory system remains idle during this phase. The system uses a synchronous clock to coordinate operations, and the next instruction fetch must only begin after write completion. A delay is observed when flag updates overlap with register writes. Analyze how the processor should coordinate the write-back sequence and infer how synchronization ensures correct completion of the instruction cycle. (5 Marks - [An/C, 2])